DISCHARGE SUMMARY – CARDIOLOGY

Patient: 67-year-old male, MRN 48-221907, admitted on 14 March 2026 with acute ST-elevation myocardial infarction (STEMI) of the inferior wall (ICD-10 I21.1). He presented with crushing retrosternal chest pain radiating to the left mandible, diaphoresis and nausea, with onset approximately 90 minutes before arrival. Blood pressure was 98/62 mmHg, heart rate 112 beats per minute and oxygen saturation 93% on room air. Electrocardiography demonstrated ST-segment elevation of 3 mm in leads II, III and aVF with reciprocal depression in leads I and aVL.

Laboratory findings: high-sensitivity troponin T 1,850 ng/L (reference <14 ng/L), NT-proBNP 4,230 pg/mL, creatinine 1.4 mg/dL (estimated glomerular filtration rate 52 mL/min/1.73 m²), potassium 4.1 mmol/L, LDL cholesterol 162 mg/dL and glycated haemoglobin (HbA1c) 7.8%. Coagulation parameters were within normal limits (INR 1.0). Chest radiography showed mild pulmonary venous congestion without pleural effusion.

Procedure: emergency percutaneous coronary intervention via right radial access. Coronary angiography revealed a thrombotic occlusion of the proximal right coronary artery (RCA) with TIMI 0 flow and a 70% stenosis of the mid left anterior descending artery. A drug-eluting everolimus stent (3.5 × 24 mm) was deployed with restoration of TIMI 3 flow. Periprocedural anticoagulation consisted of unfractionated heparin 70 IU/kg, a loading dose of ticagrelor 180 mg and acetylsalicylic acid 300 mg. Door-to-balloon time was 58 minutes. No contrast-induced nephropathy, access-site haematoma or no-reflow phenomenon occurred.

Echocardiography on day 2 showed a left ventricular ejection fraction (LVEF) of 38%, inferior wall hypokinesis, mild mitral regurgitation and no pericardial effusion. The course was complicated by transient second-degree atrioventricular block (Mobitz type I), which resolved spontaneously without pacing. Telemetry showed no ventricular arrhythmias. Cardiac rehabilitation was initiated on day 4, and the patient was mobilised without recurrence of angina or dyspnoea.

Medication at discharge: ticagrelor 90 mg twice daily for 12 months, acetylsalicylic acid 100 mg once daily, atorvastatin 80 mg at night, bisoprolol 2.5 mg once daily, ramipril 2.5 mg once daily with up-titration as tolerated, and eplerenone 25 mg once daily given the reduced ejection fraction and diabetes mellitus. Pantoprazole 40 mg once daily is prescribed for gastroprotection. Metformin was withheld for 48 hours after contrast exposure and may be resumed if the serum creatinine remains stable.

Contraindications and warnings: avoid concomitant strong CYP3A4 inhibitors, for example clarithromycin and ketoconazole, because they markedly increase ticagrelor exposure. Non-steroidal anti-inflammatory drugs are contraindicated owing to the risk of gastrointestinal haemorrhage and acute kidney injury. The patient must report melaena, haematuria, syncope or progressive dyspnoea immediately. Dual antiplatelet therapy must not be interrupted without cardiology consultation, as premature discontinuation increases the risk of stent thrombosis.

Follow-up: outpatient review in 4 weeks with repeat transthoracic echocardiography, fasting lipid profile, renal function and electrolytes. If the ejection fraction remains at or below 35% after 40 days of optimised therapy, implantation of an implantable cardioverter-defibrillator (ICD) will be evaluated. Smoking cessation, a Mediterranean diet and 150 minutes of moderate aerobic exercise per week were discussed with the patient and his wife.